

Lesson 01: Square and Cube Roots

In-class worksheet

Name: _____

Date: _____

Learning objective

Calculate and estimate square roots and cube roots.

Success criteria: I can ...

SC1 find the square root of a square number and the cube root of a cube number (for example $\sqrt{81}$ and $\sqrt[3]{125}$);

SC2 explain that a root *undoes* squaring or cubing, and check an answer by squaring or cubing it;

SC3 say which two whole numbers a root lies between, and which one it is closer to;

SC4 estimate a root to 1 decimal place and check it.

Key vocabulary

square number	A number made by multiplying a whole number by itself, e.g. $36 = 6^2$.
cube number	A number made by multiplying a whole number by itself three times, e.g. $64 = 4^3$.
square root, $\sqrt{\quad}$	$\sqrt{36}$ is the positive number that squares to give 36. $\sqrt{36} = 6$ because $6^2 = 36$.
cube root, $\sqrt[3]{\quad}$	$\sqrt[3]{64}$ is the number that cubes to give 64. $\sqrt[3]{64} = 4$ because $4^3 = 64$.
inverse operation	An operation that undoes another. Square root undoes squaring; cube root undoes cubing.
estimate	A sensible value close to the exact answer, found by reasoning and then checked.
consecutive	Next to each other in order, e.g. 6 and 7.

Starter

[Independent, 4 minutes]

S1. $8^2 = \underline{\quad}$

S3. $3^3 = \underline{\quad}$

S5. $4^3 = \underline{\quad}$

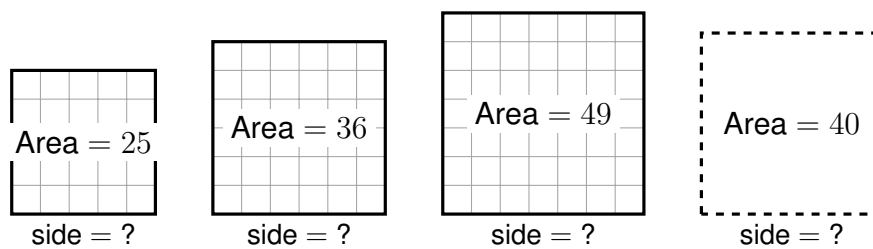
S2. $11^2 = \underline{\quad}$

S4. $10^3 = \underline{\quad}$

S6. Circle the square numbers:
20 25 36 48 81

Notice and Why

[Together]



N1. Squaring takes a side length and gives an area. What operation takes an area back to the side length? What do you notice about the side of the square with area 40?

I DO: worked examples

[Watch and annotate]

Example 1: exact roots

(a) $\sqrt{81}$ Ask: which number times *itself* is 81? $9 \times 9 = 81$. So $\sqrt{81} = 9$.

Check: $9^2 = 81$ ✓

(b) $\sqrt[3]{125}$ Ask: which number cubed is 125? $5 \times 5 \times 5 = 25 \times 5 = 125$. So $\sqrt[3]{125} = 5$.

Check: $5^3 = 125$ ✓

(c) $\sqrt[3]{-27}$ $(-3) \times (-3) \times (-3) = 9 \times (-3) = -27$. So $\sqrt[3]{-27} = -3$.

A negative number has a cube root, because *negative $\times$ negative $\times$ negative is negative*.

My notes:

Example 2: estimate $\sqrt{40}$ to 1 decimal place

Step 1: square numbers either side. $36 < 40 < 49$, so $\sqrt{36} < \sqrt{40} < \sqrt{49}$, which means $6 < \sqrt{40} < 7$.

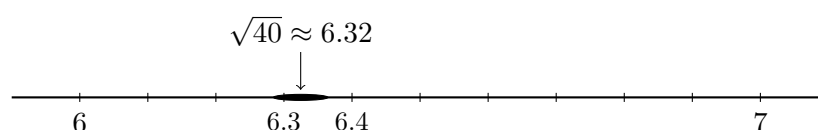
Step 2: which is it closer to? $40 - 36 = 4$ and $49 - 40 = 9$. 40 is closer to 36, so $\sqrt{40}$ is closer to 6.

Step 3: trap it between decimals. $6.3^2 = 39.69$ (too small), $6.4^2 = 40.96$ (too big). So $6.3 < \sqrt{40} < 6.4$.

Step 4: choose the closer one. $40 - 39.69 = 0.31$ and $40.96 - 40 = 0.96$. 39.69 is closer, so $\sqrt{40} \approx 6.3$ (1 d.p.).

Check with a calculator: $\sqrt{40} = 6.324... \approx 6.3$ ✓

My notes:



Example 3: estimate $\sqrt[3]{50}$

Step 1: cube numbers either side. $27 < 50 < 64$, so $3 < \sqrt[3]{50} < 4$.

Step 2: closer to? $50 - 27 = 23$ and $64 - 50 = 14$. Closer to 64, so $\sqrt[3]{50}$ is closer to 4.

Step 3: trap it. $3.6^3 = 46.656$ (too small), $3.7^3 = 50.653$ (too big).

Step 4: choose. $50 - 46.656 = 3.344$ and $50.653 - 50 = 0.653$. So $\sqrt[3]{50} \approx 3.7$ (1 d.p.). *Calculator:* $3.684 \dots \checkmark$

My notes:

WE DO: guided examples

[Together]

Support is removed a little in each question.

W1. Find (a) $\sqrt{144}$ (b) $\sqrt[3]{1000}$.

(a) $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = 144$, so $\sqrt{144} = \underline{\hspace{1cm}}$. Check: $\underline{\hspace{1cm}}^2 = 144$

(b) $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = 1000$, so $\sqrt[3]{1000} = \underline{\hspace{1cm}}$.

W2. Estimate $\sqrt{70}$ to 1 decimal place.

Square numbers either side: $\underline{\hspace{1cm}} < 70 < \underline{\hspace{1cm}}$, so $\underline{\hspace{1cm}} < \sqrt{70} < \underline{\hspace{1cm}}$.

Closer to $\underline{\hspace{1cm}}$ because $\underline{\hspace{1cm}}$.

Trap it: $\underline{\hspace{1cm}}^2 = \underline{\hspace{1cm}}$ and $\underline{\hspace{1cm}}^2 = \underline{\hspace{1cm}}$. So $\sqrt{70} \approx \underline{\hspace{1cm}}$

W3. Estimate $\sqrt[3]{20}$ to 1 decimal place. *First step:* $8 < 20 < 27$.

W4. You know $\sqrt{16} = 4$. Use this to find $\sqrt{1600}$ and $\sqrt{0.16}$. Check each answer by squaring.

Support Box (use in WE DO and YOU DO)

Square numbers	n	1	2	3	4	5	6	7	8	9	10
	n^2	1	4	9	16	25	36	49	64	81	100

Cube numbers	n	1	2	3	4	5
	n^3	1	8	27	64	125

Method for estimating a root

- Find the square (or cube) numbers either side.
- Decide which one it is closer to: compare the two differences.
- Square (or cube) decimals to trap it: $___$ ² too small, $___$ ² too big.
- Choose the decimal whose square (or cube) is closer. Check with a calculator.

Sentence stem: " $\sqrt{}$ is between $___$ and $___$ because $___ < ___ < ___$. It is closer to $___$ because $\dots$ "

YOU DO: independent practice**[Independent, 10 minutes]****How to work**

Everyone: **Foundation**, then **Core**, then **J1** (number line, after Challenge). Then move on to **Apply**. **Challenge** is optional. Check every answer by squaring or cubing. Use the Support Box above if you need it.

Foundation

- F1.** $\sqrt{25} = ______$ **F3.** $\sqrt{121} = ______$ **F5.** $\sqrt[3]{8} = ______$ **F7.** $\sqrt[3]{1000} = ______$
F2. $\sqrt{64} = ______$ **F4.** $\sqrt{196} = ______$ **F6.** $\sqrt[3]{64} = ______$ **F8.** $\sqrt{1} = ______$

Core

- C1.** Between which two consecutive whole numbers is $\sqrt{30}$? Show the square numbers you used.

- C2.** Between which two consecutive whole numbers is $\sqrt[3]{100}$? Show the cube numbers you used.

- C3.** Estimate $\sqrt{55}$ to 1 decimal place without a calculator. Then check with a calculator.

C4. Estimate $\sqrt{90}$ to 1 decimal place without a calculator. Then check with a calculator.

C5. Find $\sqrt[3]{-64}$. Show how you checked.

C6. Put in order, smallest first: $\sqrt{50}$, 7.5 , $\sqrt[3]{343}$, $\sqrt{45}$

Apply

A1. A square garden has an area of 200 m^2 . Estimate the length of one side to 1 decimal place.

A2. A cube-shaped box has a volume of 729 cm^3 . Find the length of one edge.

A3. You know $\sqrt{9} = 3$. Use this to find $\sqrt{900}$ and $\sqrt{0.09}$. Check by squaring.

A4. Find a whole number n so that $\sqrt{n}$ is between 6 and 7 *and* closer to 7. Explain how you know.

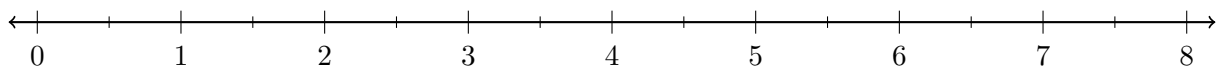
Challenge (optional)

Ch1. 64 is both a square number (8^2) and a cube number (4^3). Find the next number that is both a square number and a cube number. Show how you found it.

Ch2. Is $\sqrt{9+16}$ equal to $\sqrt{9} + \sqrt{16}$? Show working and explain.

Justify with a number line**[Independent]**

J1. Mark $\sqrt{20}$, $\sqrt[3]{20}$ and $\sqrt{50}$ on the number line as accurately as you can. For each one, write the square or cube numbers you used to decide where it goes.

**Spot the Mistake****[Together, then independent]**

For each one: **(i)** circle the *first* error, **(ii)** explain what went wrong, **(iii)** write the correct answer.

SM1. Ali: " $\sqrt{16} = 8$ because $16 \div 2 = 8$."

SM2. Bea: " $49 < 60 < 64$, so $\sqrt{60}$ is between 7 and 8. It must be in the middle, so $\sqrt{60} = 7.5$."

SM3. Cai: “ $\sqrt[3]{-8}$ has no answer, because you cannot find a root of a negative number.”

Exit Ticket**[Independent, 5 minutes, no calculator]****Exit Ticket: Lesson 01****Name:** _____**X1.** Find (a) $\sqrt{49}$ (b) $\sqrt[3]{27}$ **X2.** Between which two consecutive whole numbers is $\sqrt{85}$? Show the square numbers you used.**X3.** Estimate $\sqrt{85}$ to 1 decimal place. Show a check.**X4.** A cube has a volume of 8000 cm^3 . Find the length of one edge. Explain how you know you are right.